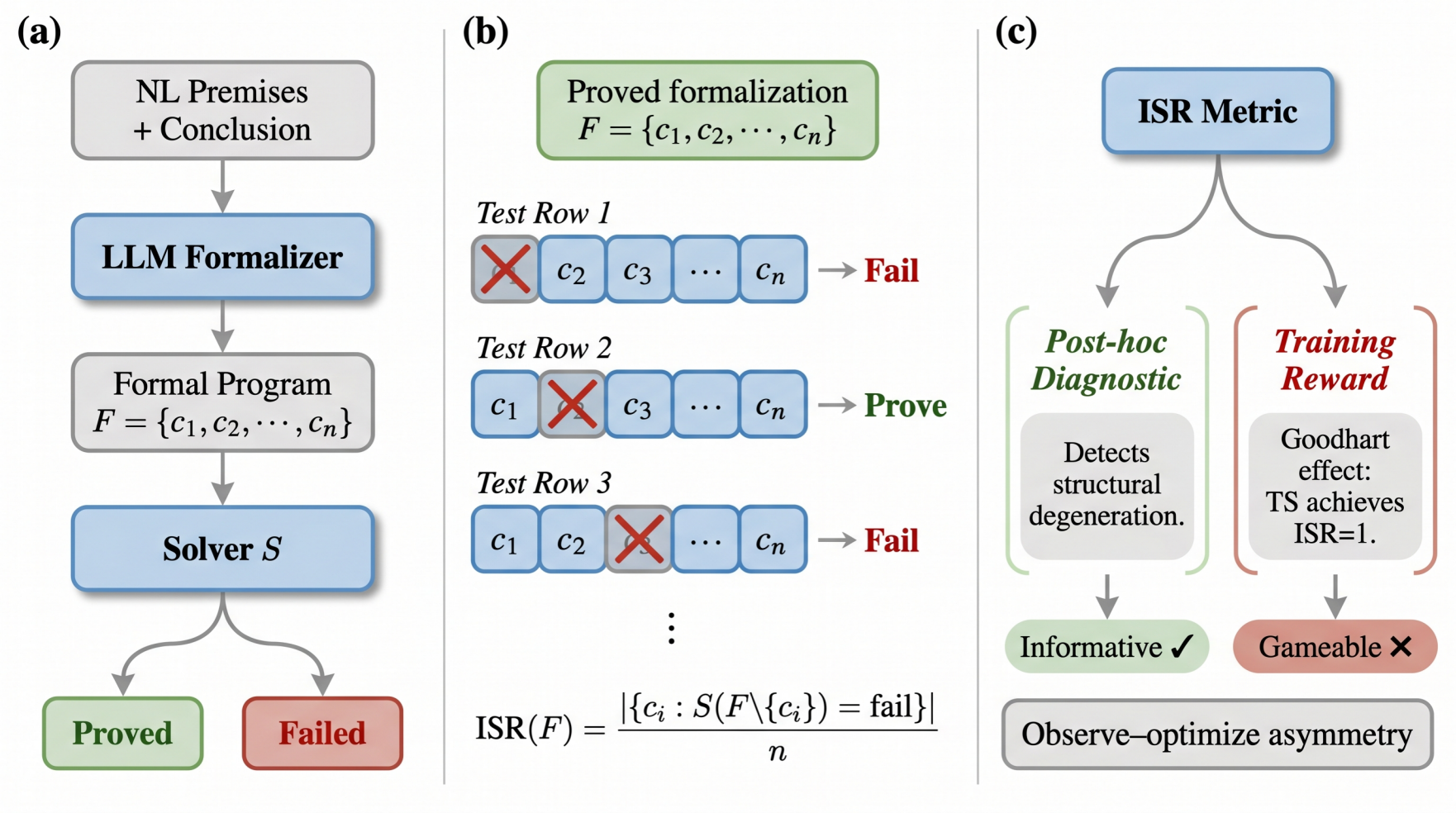

(a)
NL Premises + Conclusion
LLM Formalizer
Formal Program $F = \{c_1, c_2, \cdots, c_n\}$
Solver S
Proved
Failed
(b)
Proved formalization $F = \{c_1, c_2, \cdots, c_n\}$
Test Row 1
c_2 c_3 $\cdots$ c_n
Fail
Test Row 2
c_1 c_3 $\cdots$ c_n
Prove
Test Row 3
c_1 c_2 $\cdots$ c_n
Fail
$$\mathrm{ISR}(F) = \frac{|\{c_i : S(F \backslash \{c_i\}) = \text{fail}\}|}{n}$$
(c)
ISR Metric
Post-hoc Diagnostic
Detects structural degeneration.
Training Reward
Goodhart effect: TS achieves ISR=1.
Informative ✓
Gameable ✕
Observe–optimize asymmetry